

WEEK 1 REPORT

DATA CLEANING AND TRANSFORMATION USING POWER BI

Superstore Sales Data Import, Power Query and Data Preparation

1. Introduction

Microsoft Power BI is a business intelligence and data visualization tool used to analyze data and create meaningful reports. In this activity, I used Power BI Desktop and Power Query Editor to import, inspect, clean, and transform the Superstore Sales Dataset.

The dataset was examined to identify incorrect data types, missing values, blank or null records, unwanted spaces, and inconsistent text values. Various data transformation techniques were applied to improve the quality and usability of the dataset. Date-related information such as Year, Quarter, Month, Day, and Day of Week was also created from the Order Date column.

After completing the required transformations, the dataset was checked and loaded into the Power BI data model for further analysis and report development.

2. Objectives

The main objectives of this activity were:

- To import the Superstore Sales Dataset into Power BI.
- To understand the different views available in Power BI Desktop.
- To explore the dataset using Power Query Editor.
- To check and correct the data types of important columns.
- To identify blank, null, empty, and error values.
- To clean text data and remove unwanted spaces or characters.
- To rename columns wherever required.
- To split columns containing multiple values when necessary.
- To create date-based columns from the Order Date field.
- To prepare a clean and structured dataset for further analysis.

3. Tools and Dataset

Tool Used: Microsoft Power BI Desktop

Data Preparation Tool: Power Query Editor

Dataset: Superstore Sales Dataset

File Format: CSV

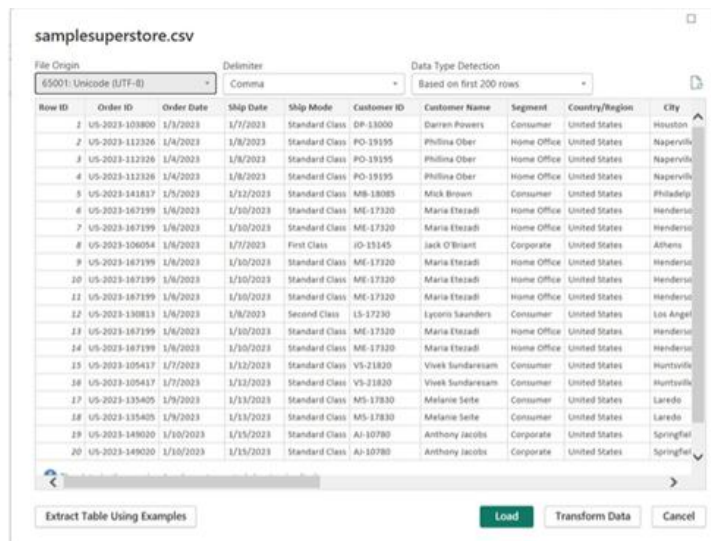
Power Query Editor was used to perform the major data cleaning and transformation operations before loading the data into the Power BI model.

4. Importing the Superstore Sales Dataset

The first step was to import the Superstore Sales CSV file into Power BI Desktop. After importing the dataset, the data was opened in Power Query Editor to examine its structure and perform the required cleaning operations.

Steps Followed

1. Opened Microsoft Power BI Desktop.
2. Selected the **Home** tab.
3. Clicked **Get Data**.
4. Selected **Text/CSV**.
5. Selected the Superstore Sales CSV file from the computer.
6. Clicked **Open**.
7. Checked the data preview.
8. Selected **Transform Data**.
9. The dataset was opened in Power Query Editor.



The screenshot shows the 'Import Data' dialog box in Power BI Desktop. The file 'samplesuperstore.csv' is selected. The 'File Origin' is '65001: Unicode (UTF-8)', the 'Delimiter' is 'Comma', and 'Data Type Detection' is 'Based on first 200 rows'. A preview of the data is shown below the settings.

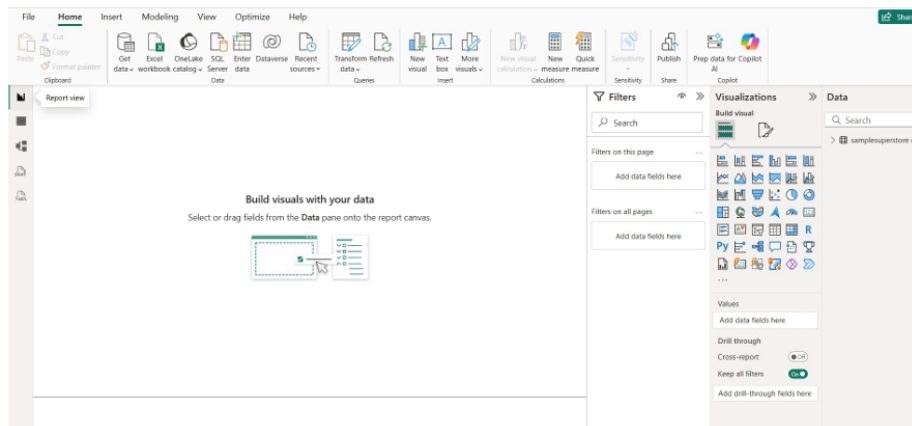
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City
1	US-2023-103800	1/3/2023	1/7/2023	Standard Class	OP-13090	Darren Powers	Consumer	United States	Houston
2	US-2023-112326	1/4/2023	1/8/2023	Standard Class	PO-19195	Phyllis Ober	Home Office	United States	Naperville
3	US-2023-112326	1/4/2023	1/8/2023	Standard Class	PO-19195	Phyllis Ober	Home Office	United States	Naperville
4	US-2023-112326	1/4/2023	1/8/2023	Standard Class	PO-19195	Phyllis Ober	Home Office	United States	Naperville
5	US-2023-141817	1/5/2023	1/12/2023	Standard Class	MB-18085	Mick Brown	Consumer	United States	Philadelph
6	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Elzacadi	Home Office	United States	Henderson
7	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Elzacadi	Home Office	United States	Henderson
8	US-2023-106054	1/6/2023	1/7/2023	First Class	JO-15145	Jack O'Brian	Corporate	United States	Athens
9	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Elzacadi	Home Office	United States	Henderson
10	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Elzacadi	Home Office	United States	Henderson
11	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Elzacadi	Home Office	United States	Henderson
12	US-2023-130813	1/6/2023	1/8/2023	Second Class	LS-17230	Lycoris Saunders	Consumer	United States	Los Angel
13	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Elzacadi	Home Office	United States	Henderson
14	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Elzacadi	Home Office	United States	Henderson
15	US-2023-105417	1/7/2023	1/12/2023	Standard Class	VS-21820	Viviek Sundaresam	Consumer	United States	Huntsville
16	US-2023-105417	1/7/2023	1/12/2023	Standard Class	VS-21820	Viviek Sundaresam	Consumer	United States	Huntsville
17	US-2023-135405	1/9/2023	1/11/2023	Standard Class	MS-17830	Melanie Sette	Consumer	United States	Laredo
18	US-2023-135405	1/9/2023	1/11/2023	Standard Class	MS-17830	Melanie Sette	Consumer	United States	Laredo
19	US-2023-149020	1/10/2023	1/15/2023	Standard Class	AI-10780	Anthony Jacobs	Corporate	United States	Springfie
20	US-2023-149020	1/10/2023	1/15/2023	Standard Class	AI-10780	Anthony Jacobs	Corporate	United States	Springfie

5. Exploring Power BI Desktop Views

Power BI Desktop provides different views for working with data and creating reports.

Report View

Report View is used to create visual reports and dashboards using charts, tables, cards, slicers, and other visualizations.



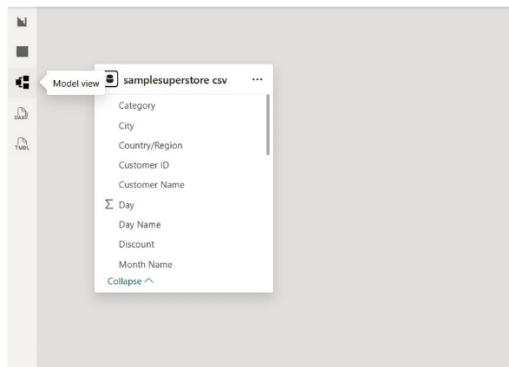
Data View

Data View allows the user to inspect the actual records in the dataset. It is useful for checking rows, columns, and values.

	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City	State/Province	Postal Code	Reg
1	S-2023-130813	06 January 2023	08 January 2023	Second Class	LS-17230	Lynco Saunders	Consumer	United States	Los Angeles	California	90049	V
57	US-2023-148614	20 January 2023	25 January 2023	Standard Class	MV-17485	Mark Van Huff	Consumer	United States	Los Angeles	California	90049	V
122	US-2023-164903	20 February 2023	24 February 2023	Standard Class	SR-20740	Steven Roelle	Home Office	United States	Los Angeles	California	90049	V
331	US-2023-102652	06 April 2023	12 April 2023	Standard Class	AY-10555	Andy Yotov	Corporate	United States	Los Angeles	California	90049	V
332	US-2023-102652	06 April 2023	12 April 2023	Standard Class	AY-10555	Andy Yotov	Corporate	United States	Los Angeles	California	90049	V
370	US-2023-120740	15 April 2023	15 April 2023	Same Day	PS-18970	Paul Stevenson	Home Office	United States	Los Angeles	California	90049	V
373	US-2023-110849	18 April 2023	23 April 2023	Standard Class	IL-15835	John Lee	Consumer	United States	Los Angeles	California	90049	V
375	US-2023-110849	18 April 2023	23 April 2023	Standard Class	IL-15835	John Lee	Consumer	United States	Los Angeles	California	90049	V
377	US-2023-110849	18 April 2023	23 April 2023	Standard Class	IL-15835	John Lee	Consumer	United States	Los Angeles	California	90049	V
723	US-2023-114125	09 July 2023	13 July 2023	Standard Class	GH-14410	Gary Hansen	Home Office	United States	Los Angeles	California	90049	V
727	US-2023-114125	09 July 2023	13 July 2023	Standard Class	GH-14410	Gary Hansen	Home Office	United States	Los Angeles	California	90049	V
1409	US-2023-101931	28 October 2023	31 October 2023	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California	90049	V
1410	US-2023-101931	28 October 2023	31 October 2023	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California	90049	V
1411	US-2023-101931	28 October 2023	31 October 2023	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California	90049	V
1422	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	V
1427	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	V
1428	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	V
1430	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	V
1431	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	V
1446	US-2023-168305	01 November 2023	08 November 2023	Standard Class	PL-18925	Paul Lucas	Home Office	United States	Los Angeles	California	90049	V
1449	US-2023-168305	01 November 2023	08 November 2023	Standard Class	PL-18925	Paul Lucas	Home Office	United States	Los Angeles	California	90049	V
1468	US-2023-153619	03 November 2023	03 November 2023	Same Day	TC-21535	Tracy Collins	Home Office	United States	Los Angeles	California	90049	V
1816	US-2023-138737	07 December 2023	10 December 2023	First Class	FP-14320	Frank Preis	Consumer	United States	Los Angeles	California	90049	V
1859	US-2023-113578	12 December 2023	15 December 2023	Second Class	KD-16345	Katherine Duoch	Consumer	United States	Los Angeles	California	90049	V
2288	US-2024-138219	23 March 2024	30 March 2024	Standard Class	BP-11095	Bert Pistole	Corporate	United States	Los Angeles	California	90049	V
3056	US-2024-121188	28 August 2024	04 September 2024	Standard Class	CB-12025	Cassandra Brandow	Consumer	United States	Los Angeles	California	90049	V

Model View

Model View displays the structure of the data model and is mainly used to manage tables and relationships.



Steps Followed

1. Opened Power BI Desktop.
2. Examined the Report View.
3. Opened Data View to inspect the records.
4. Checked Model View.
5. Reviewed the available table and fields.
6. Confirmed that the dataset was imported properly.

6. Checking and Correcting Data Types

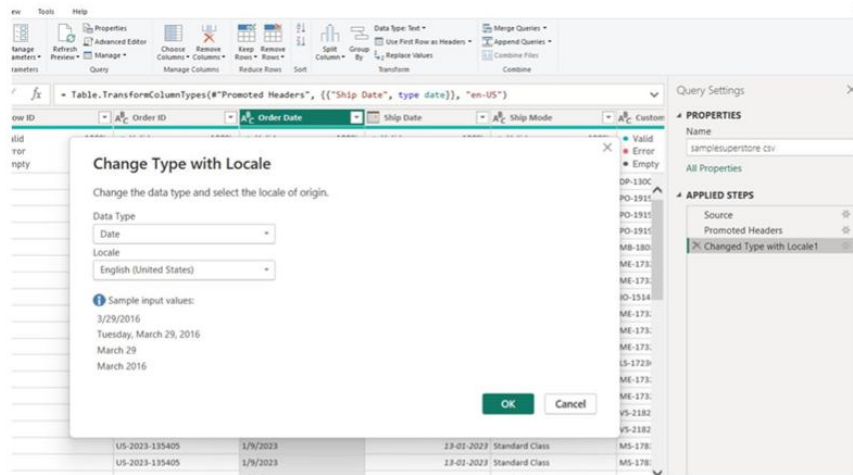
Correct data types are important for accurate data analysis and calculations. After importing the dataset, the data type of the important columns was checked in Power Query Editor.

Data Types Considered

- Text fields → **Text**
- Quantity-related fields → **Whole Number**
- Sales and Profit → **Decimal Number**
- Order Date → **Date**

Steps Followed

1. Opened the dataset in Power Query Editor.
2. Selected each important column.
3. Checked the existing data type.
4. Identified columns with incorrect or unsuitable types.
5. Used the **Data Type** option to change the required types.
6. Set text-based columns as Text.
7. Set numerical fields as Whole Number or Decimal Number.
8. Changed the Order Date column to Date format.
9. Verified the updated data types.



7. Cleaning Text Data

Text values may contain unnecessary spaces or unwanted non-printable characters. To improve consistency, text-cleaning operations were performed using the **Trim** and **Clean** options.

The main columns considered for text cleaning included:

- Category
- Sub-Category
- Segment

Steps Followed

1. Selected the required text column.
2. Opened the **Transform** tab.
3. Selected **Format**.
4. Applied the **Trim** operation.
5. Checked the values after removing extra spaces.
6. Selected **Format** → **Clean**.
7. Verified the cleaned values.
8. Repeated the same process for the required categorical columns.
9. Reviewed the final text values for consistency.

Trim removes unnecessary spaces from the beginning and end of text, while **Clean** removes non-printable characters.

9. Column Transformation

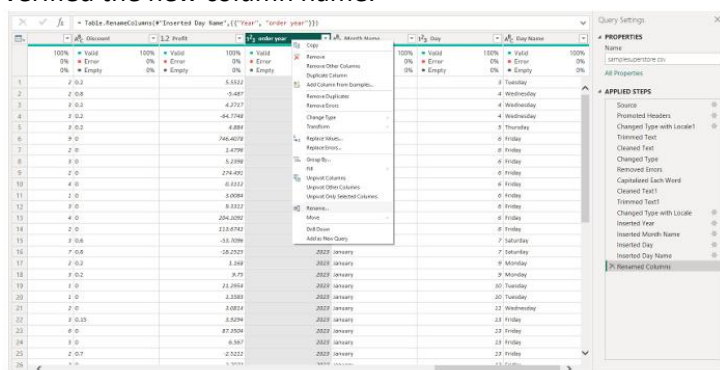
Power Query provides different transformation options that can be used to make the dataset more organized and understandable.

A. Renaming Columns

Columns were renamed wherever necessary to make their names clear and meaningful.

Steps Followed

1. Selected the required column.
2. Right-clicked on the column.
3. Selected **Rename**.
4. Entered the appropriate column name.
5. Pressed Enter.
6. Verified the new column name.



7.

B. Splitting Columns

Where a column contained more than one piece of information, the **Split Column** option was used to divide the information into separate columns.

Steps Followed

1. Selected the required column.
2. Opened the **Transform** tab.
3. Selected **Split Column**.
4. Selected the required splitting method.
5. Specified the delimiter or position.
6. Applied the transformation.
7. Checked the newly created columns.

10. Creating Date-Based Columns

The **Order Date** column was converted into the Date data type. Additional columns were created from Order Date to support time-based analysis.

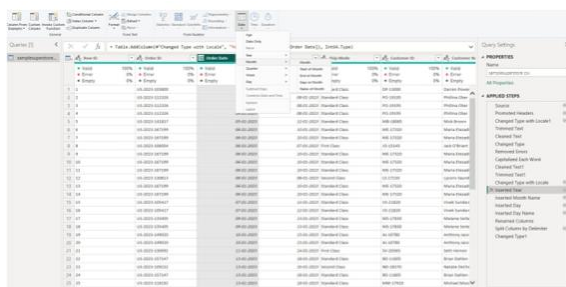
The following date attributes were created:

- Year
- Quarter
- Month
- Day
- Day of Week

A. Creating Year

1. Selected the **Order Date** column.
2. Opened the **Add Column** tab.
3. Selected **Date**.
4. Selected **Year**.
5. Chose **Year**.
6. A new Year column was generated.
7. Verified the values.

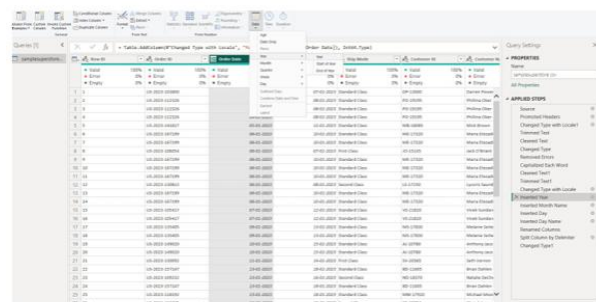
8.



B. Creating Quarter

1. Selected the Order Date column.
2. Opened **Add Column**.
3. Selected **Date**.
4. Selected **Quarter**.
5. Chose **Quarter of Year**.
6. Verified the Quarter column.

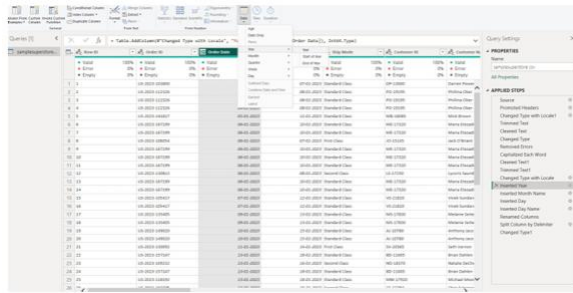
7.



C. Creating Month

1. Selected Order Date.
2. Opened the **Add Column** tab.
3. Selected **Date**.
4. Selected **Month**.

5. Selected the required month option.
6. Checked the generated Month column.



D. Creating Day

1. Selected the Order Date column.
2. Opened **Add Column**.
3. Selected **Date**.
4. Selected **Day**.
5. Chose the required Day option.
6. Verified the generated values.

E. Creating Day of Week

1. Selected the Order Date column.
2. Opened **Add Column**.
3. Selected **Date**.
4. Selected **Day**.
5. Chose **Name of Day**.
6. Verified the Day of Week column

11. Final Data Validation

After completing the cleaning and transformation process, the dataset was reviewed to ensure that all changes were applied correctly.

Final Checks Performed

- Checked the data types of important columns.
- Confirmed that Order Date was in Date format.
- Verified Year, Quarter, Month, Day, and Day of Week columns.
- Checked for blank and null values.
- Reviewed error values.
- Verified the cleaned text columns.
- Checked renamed columns.
- Reviewed the transformed columns.
- Examined the complete dataset before applying the changes.
-

The validation process ensured that the dataset was properly cleaned and ready for further analysis.

Queries [1]

Table.AddColumn(#"Changed Type with Locale", "Year", each Date.Year([Order Date]), Int64.Type)

Query Settings

Properties

Name: samplestore.csv

Applied Steps

Source

Promoted Headers

Changed Type with Locale1

Trimmed Text

Cleaned Text

Changed Type

Removed Errors

Capitalized Each Word

Cleaned Text1

Trimmed Text1

Changed Type with Locale

Inserted Year

Inserted Month Name

Inserted Day

Inserted Day Name

Renamed Columns

Split Column by Delimiter

Changed Type1

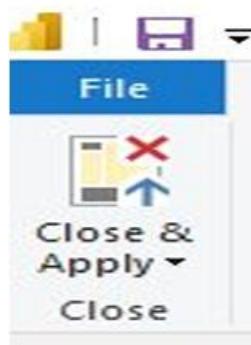
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name
1	US-2023-101800	03-01-2023	07-01-2023	Standard Class	DP-13000	Barren Power
2	US-2023-111326	04-01-2023	08-01-2023	Standard Class	PO-13195	Phyllina Ober
3	US-2023-111326	04-01-2023	08-01-2023	Standard Class	PO-13195	Phyllina Ober
4	US-2023-111326	04-01-2023	08-01-2023	Standard Class	PO-13195	Phyllina Ober
5	US-2023-141817	05-01-2023	12-01-2023	Standard Class	MB-18085	Mick Brown
6	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Ezeadi
7	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Ezeadi
8	US-2023-109094	06-01-2023	07-01-2023	First Class	JO-15145	Jack O'Brian
9	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Ezeadi
10	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Ezeadi
11	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Ezeadi
12	US-2023-130813	06-01-2023	08-01-2023	Second Class	LS-17230	Lycoris Saund
13	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Ezeadi
14	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Ezeadi
15	US-2023-109417	07-01-2023	12-01-2023	Standard Class	VS-21820	Uwek Sundari
16	US-2023-109417	07-01-2023	12-01-2023	Standard Class	VS-21820	Uwek Sundari
17	US-2023-135405	09-01-2023	13-01-2023	Standard Class	MS-17830	Melanie Sette
18	US-2023-135405	09-01-2023	13-01-2023	Standard Class	MS-17830	Melanie Sette
19	US-2023-149020	10-01-2023	15-01-2023	Standard Class	AJ-10780	Anthony Jaco
20	US-2023-149020	10-01-2023	15-01-2023	Standard Class	AJ-10780	Anthony Jaco
21	US-2023-130962	11-01-2023	14-01-2023	First Class	SV-20365	Seth Vorman
22	US-2023-157147	12-01-2023	18-01-2023	Standard Class	BD-11005	Brian Dahlen
23	US-2023-109232	13-01-2023	16-01-2023	Second Class	ND-18370	Natalia DaChu
24	US-2023-157147	12-01-2023	18-01-2023	Standard Class	BD-11005	Brian Dahlen
25	US-2023-118192	13-01-2023	18-01-2023	Standard Class	MM-17920	Michael Muso
26	US-2023-101318	14-01-2023	18-01-2023	Standard Class	ME-17320	Maria Ezeadi

12. Applying the Changes

After completing the required transformations, the changes were applied to the Power BI data model.

Steps Followed

1. Reviewed the transformed dataset.
2. Checked all applied changes.
3. Opened the **Home** tab.
4. Selected **Close & Apply**.
5. Power BI processed the transformations.
6. The cleaned dataset was loaded into the Power BI data model



13. Final Result

The Superstore Sales Dataset was successfully imported into Power BI and prepared using Power Query Editor. The required data types were checked and corrected, while text fields were cleaned using Trim and Clean operations.

Missing, blank, null, and error values were also reviewed. Required column transformations such as renaming and splitting were performed. Date attributes including Year, Quarter, Month, Day, and Day of Week were generated from the Order Date column.

Finally, the transformed dataset was validated and successfully loaded into the Power BI data model using **Close & Apply**.

14. Conclusion

In this Week 1 activity, I gained practical experience in preparing a dataset using Microsoft Power BI and Power Query Editor. I learned how to import a CSV file, explore its structure, check data types, identify missing values, and clean text fields.

I also learned how to perform basic column transformations and create useful date-based columns from an existing date field. The final validation and Close & Apply process helped me understand how cleaned data is prepared for further visualization and analysis.

Overall, this activity provided a strong foundation in data cleaning and transformation, which are essential steps for creating accurate and meaningful Power BI reports and dashboards.